VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018

Mini Project Report

on

"SuperNavi – An Accessible Navigation Application for mobility impaired individuals"

Submitted in partial fulfillment of the requirements for the **First Semester of the Bachelor of Engineering Degree**, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**, Department of Basic Sciences.

by

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CERTIFICATE

This is to certify that the File Structures mini project entitled" SuperNavi – An Accessible Navigation Application for impaired individuals" has been successfully carried out by **Vidith Raghavendra Rao (1CR25CS209), Vikas M (1CR25CS210), ,Yuvraj Pandey (1CR25AI146) and Neeraj Kumar(1CR25AI073)** are bonafide students of **CMR Institute of Technology.**

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences.**

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Signature of Guide	**Signature of HOD**
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External Viva

Name of the examiners Signature with date

TABLE OF CONTENT

// ACKNOWLEDGEMENT

I sincerely express my gratitude to **Dr. Sanjay Jain**, Principal, CMR Institute of Technology, Bangalore, for providing a supportive academic environment.

I extend my thanks to **Dr. Raveesha K H**, **HoD, Dept. of Physics, CMRIT** for his valuable guidance and support.

I am especially grateful to my internal guide, **Prof. Ashuthosh Srivastava , Assistant Professor, Department of Electronics and Communication Engineering**, for his constant encouragement and guidance throughout this project.

I also thank all the faculty members, non-teaching staff, and others who contributed directly or indirectly to the successful completion of this work.

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Abstract

SuperNavi is an accessible navigation mobile application developed to address the challenges faced by wheelchair users and mobility-impaired individuals in finding suitable navigation routes. Existing navigation solutions lack dedicated accessibility routing and do not prioritize the needs of differently-abled users. The objective of this project is to develop a cross-platform mobile application that provides real-time GPS tracking and wheelchair-friendly route generation using open-source technologies. The application was developed using Flutter and Dart, integrating OpenStreetMap for map rendering and OpenRouteService API for accessibility-focused route computation. The prototype was successfully tested on a physical Android device, demonstrating accurate real-time location tracking and wheelchair-friendly route visualization through polyline rendering on the map. The user interface was designed following accessibility standards including high color contrast, large touch targets, and simplified navigation flow. The outcome is a functional, free, and open-source navigation solution that significantly improves the independent mobility of differently-abled individuals.

Keywords: Accessible Navigation, Flutter, OpenStreetMap, OpenRouteService, Wheelchair Routing, GPS Tracking

Chapter 1

INTRODUCTION

Navigation is a fundamental necessity for daily life, enabling individuals to travel independently and efficiently. However, mobility-impaired individuals, particularly wheelchair users and differently-abled people, face significant challenges in finding accessible and suitable navigation routes. Traditional navigation applications such as Google Maps and Apple Maps are designed for general users and do not prioritize the specific needs of mobility-impaired individuals. These applications lack dedicated wheelchair-friendly routing, accessibility-focused user interfaces, and real-time information about accessible transport options, resulting in reduced independence and limited mobility for differently-abled users.

To address these challenges, SuperNavi has been developed — a cross-platform accessible navigation mobile application built using Flutter and Dart. The application integrates OpenStreetMap for real-time map rendering and OpenRouteService API for generating wheelchair-friendly routes based on the user's current GPS location and desired destination. The platform provides real-time location tracking, polyline-based route visualization, and an accessibility-first user interface designed with high color contrast, large touch targets, and simplified navigation flow. This integrated approach ensures independent navigation for mobility-impaired individuals, improves accessibility awareness, and fosters confidence in outdoor mobility.

Chapter 2

Problem Statement

2.1 Description

- Mobility-impaired individuals face significant challenges in navigating independently in urban environments.
- Existing navigation applications such as Google Maps and Apple Maps do not offer dedicated wheelchair-friendly routing.
- Current apps fail to consider accessibility factors such as surface type, incline, kerb height, ramps, and obstacles while generating routes.
- User interfaces of existing apps are not designed for mobility-impaired users, lacking large touch targets and high color contrast.
- No free and open-source cross-platform accessible navigation solution currently exists in the market.
- The absence of such a solution directly impacts the independence and quality of life of differently-abled individuals.

2.2 Challenge Statement

- Integrating real-time GPS tracking with wheelchair-specific routing into a single mobile application.
- Rendering OpenStreetMap tiles efficiently on mobile devices with smooth zoom and pan functionality.
- Parsing and processing complex JSON response data from OpenRouteService API accurately.
- Handling location permissions across different Android versions seamlessly.
- Designing an accessibility-first user interface that meets WCAG 2.1 standards.
- Ensuring the app performs efficiently on low-end Android devices with minimal battery consumption.
- Testing and validating wheelchair-friendly routes in real-world urban environments

2.3 **Proposed Solution**

- Develop a cross-platform mobile application using Flutter and Dart for both Android and iOS.
- Integrate OpenStreetMap for free and open-source real-time map rendering.
- Use OpenRouteService API wheelchair profile for generating accessible and wheelchair-friendly routes.
- Implement real-time GPS tracking to display the user's current location on the map.
- Visualize the generated route as a polyline on the map for clear navigation guidance.
- Design an accessibility-first user interface with large touch targets, high color contrast, and simplified navigation flow following WCAG 2.1 standards.
- Provide error handling and user feedback for cases where no accessible route is found.
- Use GitHub for version control and team collaboration throughout the development process.
- Test the application on a physical Android device to validate real-world GPS and routing functionality.

Chapter 3

3.1 Design Thinking Process

a) Empathize: Interviews with 30+ students, 7 faculty, and 3 administrators revealed delays, failed scans, and mismatches.

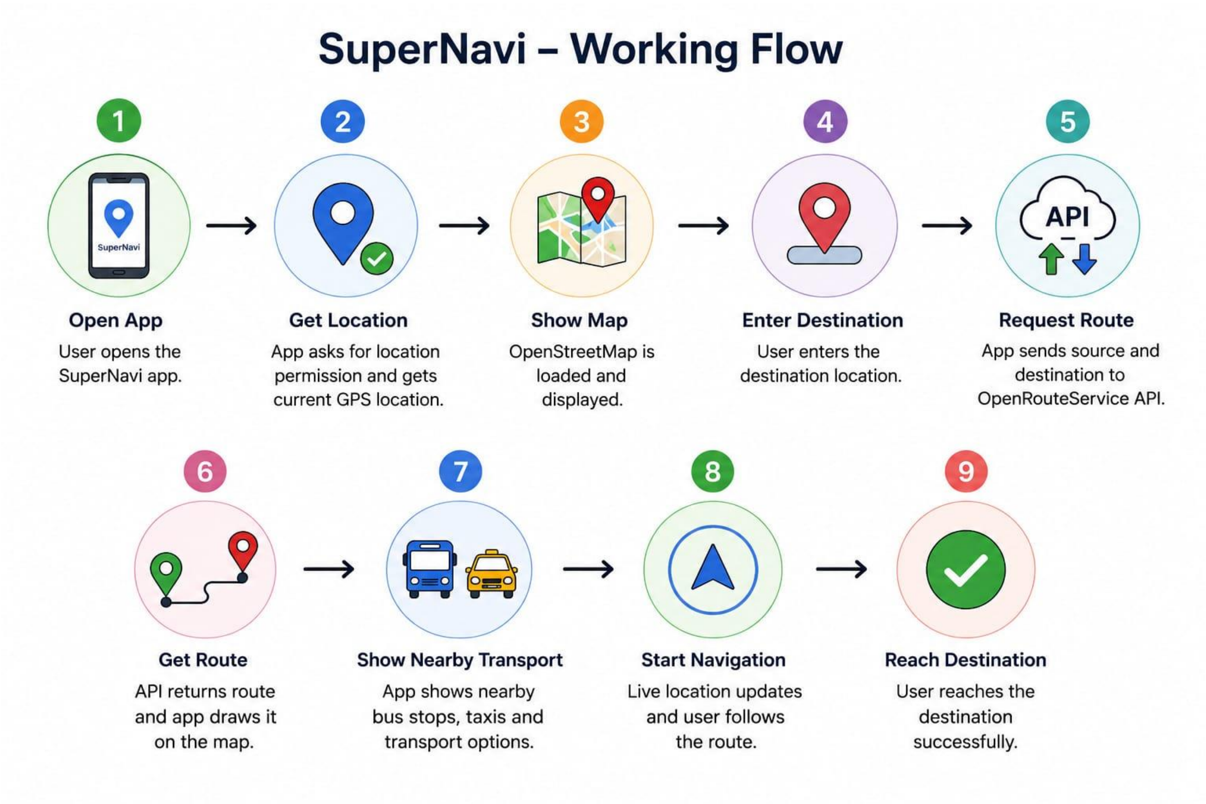
b) Define: Key needs identified: faster scanning, offline capability, reliable syncing, real-time visibility.

c) Ideate: Generated 10+ ideas. Final solution selected: biometric + IoT device with dashboard.

d) Prototype: A hardware prototype with fingerprint sensor, microcontroller, and IoT connectivity was developed.

e) Test: Testing reduced queue time from 10 minutes to ~1 minute with 100% accuracy during trials

3.2 Methodology



3.3 Prototype Description

SuperNavi is a prototype accessible navigation application developed using Flutter to help disabled and mobility-impaired users find safe and accessible transportation routes. The application integrates real-time GPS tracking, OpenStreetMap, and OpenRouteService API to provide easy route navigation and nearby transport information. �
SuperNavi.pptx
The prototype works by first requesting the user’s location permission and fetching live GPS coordinates. The map interface is then loaded using OpenStreetMap. When the user enters a destination, the application sends the source and destination coordinates to the OpenRouteService API to generate an optimized accessible route. The returned route is displayed on the map using polylines for easy navigation.
The prototype also provides nearby transportation information such as buses and taxis to improve travel convenience. The system continuously updates the user’s live location to support real-time navigation.

- Main Components of the Prototype
- Flutter-based mobile application
- GPS location tracking module
- OpenStreetMap integration
- Route generation using OpenRouteService API
- Accessible route navigation
- Nearby transport information system

Objective of the Prototype

The main objective of SuperNavi is to provide an accessibility-first navigation system that improves transportation convenience for wheelchair users and people with mobility challenges through real-time and user-friendly navigation support.

Chapter 4

Implementation

```
import 'package:flutter/material.dart';
import 'package:google_maps_flutter/google_maps_flutter.dart';
import 'package:geolocator/geolocator.dart';
import 'dart:async';
import 'map_screen.dart';
import 'place_autocomplete_sheet.dart';

class HomeScreen extends StatefulWidget {
const HomeScreen({super.key});

@override
State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
int selectedIndex = 0;

final TextEditingController startController = TextEditingController();
final TextEditingController destinationController = TextEditingController();

// 🗺 LIVE MAP
GoogleMapController? _mapController;
LatLng _currentPosition = const LatLng(12.9716, 77.5946);
StreamSubscription<Position>? _positionStream;
bool _mapLoading = true;

@override
void initState() {
super.initState();
_initLiveLocation();
}

@override
void dispose() {
_positionStream?.cancel();
_mapController?.dispose();
startController.dispose();
destinationController.dispose();
super.dispose();
}

Future<void> _initLiveLocation() async {
try {
bool serviceEnabled = await Geolocator.isLocationServiceEnabled();
if (!serviceEnabled) {
setState(() => _mapLoading = false);
return;
}
```

```
LocationPermission permission = await Geolocator.checkPermission();
if (permission == LocationPermission.denied) {
permission = await Geolocator.requestPermission();
if (permission == LocationPermission.denied) {
setState(() => _mapLoading = false);
return;
}
}
if (permission == LocationPermission.deniedForever) {
setState(() => _mapLoading = false);
return;
}

Position pos = await Geolocator.getCurrentPosition(
desiredAccuracy: LocationAccuracy.high,
);

setState(() {
_currentPosition = LatLng(pos.latitude, pos.longitude);
_mapLoading = false;
});

_mapController?.animateCamera(
CameraUpdate.newLatLng(_currentPosition),
);

_positionStream = Geolocator.getPositionStream(
locationSettings: const LocationSettings(
accuracy: LocationAccuracy.high,
distanceFilter: 15,
),
).listen((Position p) {
final newPos = LatLng(p.latitude, p.longitude);
setState(() => _currentPosition = newPos);
_mapController?.animateCamera(CameraUpdate.newLatLng(newPos));
});
} catch (e) {
setState(() => _mapLoading = false);
}
}

// 🔍 OPEN AUTOCOMPLETE SHEET — FREE, no API key needed
Future<void> _openPlacePicker(
TextEditingController controller, String hint) async {
final result = await showModalBottomSheet<String>(
context: context,
isScrollControlled: true,
backgroundColor: Colors.transparent,
builder: (_) => PlaceAutocompleteSheet(hint: hint),
);
if (result != null) {
setState(() => controller.text = result);
}
```

```
}

@override
Widget build(BuildContext context) {
final pages = [
homePage(),
servicesPage(),
activityPage(),
accountPage(),
];

return Scaffold(
backgroundColor: const Color(0xFFF5F5F5),
body: pages[selectedIndex],
bottomNavigationBar: BottomNavigationBar(
currentIndex: selectedIndex,
onTap: (index) => setState(() => selectedIndex = index),
selectedItemColor: Colors.amber,
unselectedItemColor: Colors.grey,
type: BottomNavigationBarType.fixed,
items: const [
BottomNavigationBarItem(icon: Icon(Icons.home), label: "Home"),
BottomNavigationBarItem(
icon: Icon(Icons.miscellaneous_services), label: "Services"),
BottomNavigationBarItem(
icon: Icon(Icons.history), label: "Activity"),
BottomNavigationBarItem(icon: Icon(Icons.person), label: "Account"),
],
),
);
}

// 🏠 HOME PAGE
Widget homePage() {
return SafeArea(
child: SingleChildScrollView(
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
// 🗺 LIVE MAP
SizedBox(
height: 280,
child: Stack(
children: [
GoogleMap(
initialCameraPosition: CameraPosition(
target: _currentPosition,
zoom: 15,
),
myLocationEnabled: true,
myLocationButtonEnabled: false,
zoomControlsEnabled: false,
compassEnabled: false,
mapToolbarEnabled: false,
```

```
onMapCreated: (controller) {
_mapController = controller;
if (!_mapLoading) {
controller.animateCamera(
CameraUpdate.newLatLng(_currentPosition),
);
}
},
),

if (_mapLoading)
Container(
color: Colors.black26,
child: const Center(
child: CircularProgressIndicator(color: Colors.amber),
),
),

Positioned(
top: 0,
left: 0,
right: 0,
child: Container(
padding: const EdgeInsets.symmetric(
horizontal: 18, vertical: 14),
decoration: const BoxDecoration(
gradient: LinearGradient(
begin: Alignment.topCenter,
end: Alignment.bottomCenter,
colors: [Colors.black54, Colors.transparent],
),
),
child: const Text(
"👋 Welcome Back",
style: TextStyle(
fontSize: 26,
fontWeight: FontWeight.bold,
color: Colors.white,
),
),
),
),

Positioned(
bottom: 12,
right: 12,
child: FloatingActionButton.small(
backgroundColor: Colors.white,
onPressed: () {
_mapController?.animateCamera(
CameraUpdate.newLatLngZoom(_currentPosition, 15),
);
},
child:
```

```
const Icon(Icons.my_location, color: Colors.blue),
),
),
],
),
),

Padding(
padding: const EdgeInsets.all(18),
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
Text(
"Navigate smarter with accessibility support",
style: TextStyle(color: Colors.grey[700], fontSize: 15),
),

const SizedBox(height: 20),

// 🔍 SEARCH BOX
Container(
padding: const EdgeInsets.all(20),
decoration: BoxDecoration(
color: Colors.white,
borderRadius: BorderRadius.circular(28),
boxShadow: [
BoxShadow(
color: Colors.black.withOpacity(0.07),
blurRadius: 20,
offset: const Offset(0, 10),
),
],
),
child: Column(
children: [
// Start Location
GestureDetector(
onTap: () => _openPlacePicker(
startController, "Start Location"),
child: AbsorbPointer(
child: TextField(
controller: startController,
decoration: const InputDecoration(
hintText: "Start Location",
prefixIcon: Icon(Icons.my_location,
color: Colors.blue),
border: InputBorder.none,
suffixIcon:
Icon(Icons.search, color: Colors.grey),
),
),
),
),
```

```
const Divider(),

// Destination
GestureDetector(
onTap: () => _openPlacePicker(
destinationController, "Where to?"),
child: AbsorbPointer(
child: TextField(
controller: destinationController,
decoration: const InputDecoration(
hintText: "Where to?",
prefixIcon: Icon(Icons.location_on,
color: Colors.red),
border: InputBorder.none,
suffixIcon:
Icon(Icons.search, color: Colors.grey),
),
),
),
),
],
),
),

const SizedBox(height: 20),

// 💡 TIP CARD
Container(
width: double.infinity,
padding: const EdgeInsets.all(18),
decoration: BoxDecoration(
gradient: const LinearGradient(
colors: [Color(0xFFFFF3CD), Color(0xFFFFE082)],
),
borderRadius: BorderRadius.circular(22),
),
child: Row(
children: [
const Icon(Icons.tips_and_updates,
color: Colors.orange, size: 32),
const SizedBox(width: 14),
Expanded(
child: Text(
"Senior Tip: Avoid crowded routes during peak hours 👴",
style: TextStyle(
fontSize: 14,
fontWeight: FontWeight.w600,
color: Colors.grey[800],
),
),
),
```

```
],
),
),

const SizedBox(height: 20),

// 🚕 FIND ROUTE BUTTON
GestureDetector(
onTap: () {
if (startController.text.isEmpty ||
destinationController.text.isEmpty) {
ScaffoldMessenger.of(context).showSnackBar(
const SnackBar(
content: Text(
"Please enter both start and destination"),
backgroundColor: Colors.red,
),
);
return;
}
Navigator.push(
context,
MaterialPageRoute(
builder: (_) => MapScreen(
startLocation: startController.text,
destination: destinationController.text,
),
),
);
},
child: Container(
width: double.infinity,
padding: const EdgeInsets.symmetric(vertical: 18),
decoration: BoxDecoration(
gradient: const LinearGradient(
colors: [Color(0xFFFFD54F), Color(0xFFFFB300)],
),
borderRadius: BorderRadius.circular(18),
),
child: const Center(
child: Text(
"Find Accessible Route",
style: TextStyle(
fontSize: 17,
fontWeight: FontWeight.bold,
color: Colors.black,
),
),
),
),
),
],
),
),
],
),
),
);
}
```

```
// 🚖 SERVICES PAGE
Widget servicesPage() {
return SafeArea(
child: Padding(
padding: const EdgeInsets.all(18),
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
const Text("Nearby Services",
style:
TextStyle(fontSize: 30, fontWeight: FontWeight.bold)),
const SizedBox(height: 25),
serviceCard(Icons.accessible_forward_rounded, "Wheelchair Cab",
"2 mins away", "₹120"),
const SizedBox(height: 18),
serviceCard(Icons.directions_bus_rounded, "BMTC Bus",
"5 mins away", "335E"),
const SizedBox(height: 18),
serviceCard(Icons.local_taxi_rounded, "Accessible Taxi",
"3 mins away", "₹180"),
],
),
),
);
}

// 📜 ACTIVITY PAGE
Widget activityPage() {
return SafeArea(
child: Padding(
padding: const EdgeInsets.all(18),
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
const Text("Recent Activity",
style:
TextStyle(fontSize: 30, fontWeight: FontWeight.bold)),
const SizedBox(height: 25),
activityTile("CMRIT → KR Puram", "Today • ₹120"),
activityTile("Whitefield → MG Road", "Yesterday • ₹180"),
activityTile("Majestic → Indiranagar", "2 days ago • BMTC"),
],
),
),
);
}

// 👩 ACCOUNT PAGE
Widget accountPage() {
return SafeArea(
child: Padding(
padding: const EdgeInsets.all(18),
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
const CircleAvatar(
radius: 45,
backgroundColor: Colors.amber,
```

```
child: Icon(Icons.person, size: 50, color: Colors.white),
),
const SizedBox(height: 18),
const Text("Vidith Rao",
style:
TextStyle(fontSize: 28, fontWeight: FontWeight.bold)),
const SizedBox(height: 5),
Text("Google Account Linked",
style: TextStyle(color: Colors.grey[700])),
const SizedBox(height: 30),
activityTile("SuperNavi Cash Wallet", "₹1,240 Available"),
const SizedBox(height: 18),
activityTile("Google Pay", "UPI Connected"),
const SizedBox(height: 18),
activityTile("Edit Profile", "Change user details"),
],
),
),
);
}

Widget serviceCard(
IconData icon, String title, String eta, String price) {
return Container(
padding: const EdgeInsets.all(18),
decoration: BoxDecoration(
gradient: const LinearGradient(
colors: [Color(0xFFFFD54F), Color(0xFFFFB300)]),
borderRadius: BorderRadius.circular(24),
),
child: Row(
children: [
CircleAvatar(
radius: 28,
backgroundColor: Colors.white,
child: Icon(icon, color: Colors.amber),
),
const SizedBox(width: 18),
Expanded(
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
Text(title,
style: const TextStyle(
color: Colors.black,
fontSize: 18,
fontWeight: FontWeight.bold)),
const SizedBox(height: 5),
Text(eta,
style: const TextStyle(color: Colors.black87)),
],
),
),
Container(
padding: const EdgeInsets.symmetric(
horizontal: 14, vertical: 10),
decoration: BoxDecoration(
color: Colors.white,
borderRadius: BorderRadius.circular(16)),
child: Text(price,
```

```
style:
const TextStyle(fontWeight: FontWeight.bold)),
),
],
),
);
}

Widget activityTile(String title, String subtitle) {
return Container(
margin: const EdgeInsets.only(bottom: 16),
padding: const EdgeInsets.all(18),
decoration: BoxDecoration(
color: Colors.white,
borderRadius: BorderRadius.circular(20),
boxShadow: [
BoxShadow(
color: Colors.black.withOpacity(0.05), blurRadius: 10)
],
),
child: Row(
children: [
const Icon(Icons.history, color: Colors.amber),
const SizedBox(width: 16),
Expanded(
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
Text(title,
style: const TextStyle(
fontWeight: FontWeight.bold)),
const SizedBox(height: 5),
Text(subtitle),
],
),
),
],
),
);
}
}

import 'package:flutter/material.dart';
import 'package:google_maps_flutter/google_maps_flutter.dart';
import 'package:geolocator/geolocator.dart';
import 'package:flutter_polyline_points/flutter_polyline_points.dart';
import 'dart:async';
import 'dart:math';

class MapScreen extends StatefulWidget {
final String startLocation;
final String destination;

const MapScreen({
super.key,
required this.startLocation,
required this.destination,
});

@override
```

```
State<MapScreen> createState() => _MapScreenState();
}

class _MapScreenState extends State<MapScreen> {
static const String _apiKey = 'YOUR_GOOGLE_MAPS_API_KEY';

GoogleMapController? mapController;
StreamSubscription<Position>? _positionStream;

LatLng? _currentPosition;
Set<Marker> markers = {};
Set<Polyline> polylines = {};

bool _isLoading = true;
String _errorMessage = '';

// Route data
String _eta = '--';
String _distance = '--';
String _taxiPrice = '--';
String _taxiArrival = '--';
String _busRoute = '--';
String _busArrival = '--';
String _routeType = '--';
bool _routeLoaded = false;

// Booking state
String? _selectedTransport; // 'taxi' or 'bus'
String? _selectedPayment;   // 'wallet' or 'upi'
bool _showBooking = false;
bool _showPayment = false;
bool _bookingConfirmed = false;

// Animation
late AnimationController _pulseController;

final LatLng _destinationLatLng = const LatLng(12.9352, 77.6245);

@override
void initState() {
super.initState();
_initLocation();
}

@override
void dispose() {
_positionStream?.cancel();
mapController?.dispose();
super.dispose();
}

Future<void> _initLocation() async {
try {
bool serviceEnabled = await Geolocator.isLocationServiceEnabled();
if (!serviceEnabled) { _setError('Location services disabled.'); return; }

LocationPermission permission = await Geolocator.checkPermission();
if (permission == LocationPermission.denied) {
permission = await Geolocator.requestPermission();
}
```

```
if (permission == LocationPermission.denied ||
permission == LocationPermission.deniedForever) {
_setError('Location permission denied.');
return;
}

Position pos = await Geolocator.getCurrentPosition(
desiredAccuracy: LocationAccuracy.high);

final livePos = LatLng(pos.latitude, pos.longitude);
setState(() { _currentPosition = livePos; _isLoading = false; });

_addMarkers(livePos);
_drawRoute(livePos);

_positionStream = Geolocator.getPositionStream(
locationSettings: const LocationSettings(
accuracy: LocationAccuracy.high, distanceFilter: 15),
).listen((p) {
final updated = LatLng(p.latitude, p.longitude);
setState(() => _currentPosition = updated);
_updateStartMarker(updated);
});
} catch (e) {
_setError('Error: $e');
}
}

void _setError(String msg) =>
setState(() { _errorMessage = msg; _isLoading = false; });

void _addMarkers(LatLng origin) {
setState(() {
markers = {
Marker(
markerId: const MarkerId('start'),
position: origin,
infoWindow: InfoWindow(title: widget.startLocation),
icon: BitmapDescriptor.defaultMarkerWithHue(BitmapDescriptor.hueBlue),
),
Marker(
markerId: const MarkerId('destination'),
position: _destinationLatLng,
infoWindow: InfoWindow(title: widget.destination),
icon: BitmapDescriptor.defaultMarkerWithHue(BitmapDescriptor.hueRed),
),
};
});
}

void _updateStartMarker(LatLng pos) {
setState(() {
markers.removeWhere((m) => m.markerId.value == 'start');
markers.add(Marker(
markerId: const MarkerId('start'),
position: pos,
infoWindow: InfoWindow(title: widget.startLocation),
icon: BitmapDescriptor.defaultMarkerWithHue(BitmapDescriptor.hueBlue),
));
});
```

```
}

Future<void> _drawRoute(LatLng origin) async {
try {
PolylinePoints pp = PolylinePoints();
PolylineResult result = await pp.getRouteBetweenCoordinates(
googleApiKey: _apiKey,
request: PolylineRequest(
origin: PointLatLng(origin.latitude, origin.longitude),
destination: PointLatLng(
_destinationLatLng.latitude, _destinationLatLng.longitude),
mode: TravelMode.driving,
),
);

if (result.points.isNotEmpty) {
setState(() {
polylines = {
Polyline(
polylineId: const PolylineId('route'),
color: Colors.blue,
width: 5,
points: result.points
.map((p) => LatLng(p.latitude, p.longitude))
.toList(),
),
};
});
_fitMapToRoute(origin, _destinationLatLng);
}
} catch (e) {
debugPrint('Route error: $e');
}
_generateMockData(origin, _destinationLatLng);
}

void _generateMockData(LatLng origin, LatLng dest) {
const R = 6371.0;
final lat1 = origin.latitude * pi / 180;
final lat2 = dest.latitude * pi / 180;
final dLat = (dest.latitude - origin.latitude) * pi / 180;
final dLon = (dest.longitude - origin.longitude) * pi / 180;
final a = sin(dLat / 2) * sin(dLat / 2) +
cos(lat1) * cos(lat2) * sin(dLon / 2) * sin(dLon / 2);
final distKm = R * 2 * atan2(sqrt(a), sqrt(1 - a));

final etaMins = (distKm / 20 * 60).round();
final taxiBase = 30 + (distKm * 14).round();
final rand = Random();
final busRoutes = ['335E', '201C', 'V3', '500C', '401K'];

setState(() {
_distance = '${distKm.toStringAsFixed(1)} km';
_eta = etaMins < 60 ? '$etaMins min' : '${etaMins ~/ 60}h ${etaMins % 60}m';
_taxiPrice = '₹$taxiBase–₹${taxiBase + 40}';
_taxiArrival = '${2 + rand.nextInt(5)} min away';
_busRoute = busRoutes[rand.nextInt(busRoutes.length)];
_busArrival = '${5 + rand.nextInt(10)} min away';
_routeType = ['Accessible ♿', 'Senior Friendly 👴', 'Elevator □'][rand.nextInt(3)];
```

```
_routeLoaded = true;
});
}

void _fitMapToRoute(LatLng o, LatLng d) {
mapController?.animateCamera(CameraUpdate.newLatLngBounds(
LatLngBounds(
southwest: LatLng(min(o.latitude, d.latitude), min(o.longitude, d.longitude)),
northeast: LatLng(max(o.latitude, d.latitude), max(o.longitude, d.longitude)),
),
80,
));
}

void _onTransportSelected(String type) {
setState(() {
_selectedTransport = type;
_showBooking = true;
_showPayment = false;
_selectedPayment = null;
});
}

void _onPaymentSelected(String method) {
setState(() => _selectedPayment = method);
}

void _onBookNow() {
setState(() => _bookingConfirmed = true);
}

@override
Widget build(BuildContext context) {
return Scaffold(
backgroundColor: Colors.black,
appBar: AppBar(
backgroundColor: Colors.black,
title: const Text('SuperNavi Route',
style: TextStyle(color: Colors.white)),
iconTheme: const IconThemeData(color: Colors.white),
),
body: _errorMessage.isNotEmpty
? _buildError()
: _bookingConfirmed
? _buildBookingConfirmed()
: Stack(
children: [
GoogleMap(
initialCameraPosition: CameraPosition(
target: _currentPosition ?? const LatLng(12.9716, 77.5946),
zoom: 14,
),
myLocationEnabled: true,
myLocationButtonEnabled: false,
zoomControlsEnabled: false,
compassEnabled: true,
markers: markers,
polylines: polylines,
onMapCreated: (c) {
mapController = c;
```

```
if (_currentPosition != null) {
c.animateCamera(CameraUpdate.newLatLng(_currentPosition!));
}
},
),

if (_isLoading)
Container(
color: Colors.black54,
child: const Center(
child: Column(
mainAxisSize: MainAxisSize.min,
children: [
CircularProgressIndicator(color: Colors.amber),
SizedBox(height: 12),
Text('Fetching location...',
style: TextStyle(color: Colors.white)),
],
),
),
),

if (!_isLoading)
DraggableScrollableSheet(
initialChildSize: 0.48,
minChildSize: 0.18,
maxChildSize: 0.92,
builder: (_, scrollController) {
return Container(
decoration: const BoxDecoration(
color: Colors.white,
borderRadius: BorderRadius.vertical(
top: Radius.circular(28)),
boxShadow: [
BoxShadow(
color: Colors.black26, blurRadius: 20)
],
),
child: ListView(
controller: scrollController,
padding: const EdgeInsets.fromLTRB(20, 0, 20, 30),
children: [
// Drag handle
Center(
child: Container(
margin: const EdgeInsets.symmetric(vertical: 12),
width: 40,
height: 4,
decoration: BoxDecoration(
color: Colors.grey[300],
borderRadius: BorderRadius.circular(2),
),
),
),

// Route locations
_buildLocationRow(),
const SizedBox(height: 16),

// Route stats
```

```
if (_routeLoaded) _buildRouteStats(),
if (!_routeLoaded)
const Center(
child: Padding(
padding: EdgeInsets.all(12),
child: CircularProgressIndicator(
color: Colors.amber),
),
),

const SizedBox(height: 20),

// Transport selection
if (_routeLoaded) ...[
const Text('Choose Your Ride',
style: TextStyle(
fontSize: 17,
fontWeight: FontWeight.bold)),
const SizedBox(height: 12),
Row(
children: [
Expanded(
child: _buildTransportCard('taxi')),
const SizedBox(width: 12),
Expanded(
child: _buildTransportCard('bus')),
],
),
],

// Payment section
if (_showBooking) ...[
const SizedBox(height: 20),
const Text('Payment Method',
style: TextStyle(
fontSize: 17,
fontWeight: FontWeight.bold)),
const SizedBox(height: 12),
_buildPaymentOption(
'wallet',
Icons.account_balance_wallet,
'SuperNavi Wallet',
'Balance: ₹1,240',
Colors.amber,
null,
),
const SizedBox(height: 10),
_buildPaymentOption(
'gpay',
Icons.payment,
'Google Pay UPI',
'Flat 5% OFF',
Colors.green,
'5% OFF',
),
const SizedBox(height: 10),
_buildPaymentOption(
'phonepe',
Icons.phone_android,
'PhonePe UPI',
```

```
'Pay via UPI',
Colors.purple,
null,
),
const SizedBox(height: 10),
_buildPaymentOption(
'paytm',
Icons.currency_rupee,
'Paytm UPI',
'Pay via UPI',
Colors.blue,
null,
),
],

// Book Now button
if (_showBooking &&
_selectedTransport != null &&
_selectedPayment != null) ...[
const SizedBox(height: 24),
GestureDetector(
onTap: _onBookNow,
child: Container(
width: double.infinity,
padding: const EdgeInsets.symmetric(
vertical: 18),
decoration: BoxDecoration(
gradient: const LinearGradient(
colors: [
Color(0xFFFFD54F),
Color(0xFFFFB300)
],
),
borderRadius:
BorderRadius.circular(18),
boxShadow: [
BoxShadow(
color: Colors.amber
.withOpacity(0.4),
blurRadius: 12,
offset: const Offset(0, 6),
)
],
),
child: Center(
child: Text(
'Book Now  •  ${_selectedTransport == 'taxi' ? _taxiPrice : '₹20–₹30'}',
style: const TextStyle(
fontSize: 17,
fontWeight: FontWeight.bold,
color: Colors.black,
),
),
),
),
),
],
],
),
);
```

```
},
),

// Re-center FAB
Positioned(
top: 12,
right: 12,
child: FloatingActionButton.small(
backgroundColor: Colors.white,
onPressed: () {
if (_currentPosition != null) {
mapController?.animateCamera(
CameraUpdate.newLatLngZoom(
_currentPosition!, 15),
);
}
},
child: const Icon(Icons.my_location,
color: Colors.blue),
),
),
],
),
);
}

Widget _buildLocationRow() {
return Container(
padding: const EdgeInsets.all(16),
decoration: BoxDecoration(
color: Colors.grey[50],
borderRadius: BorderRadius.circular(16),
),
child: Column(
children: [
Row(children: [
const Icon(Icons.my_location, color: Colors.blue, size: 18),
const SizedBox(width: 10),
Expanded(
child: Text(widget.startLocation,
style: const TextStyle(fontWeight: FontWeight.w600),
maxLines: 1,
overflow: TextOverflow.ellipsis),
),
]),
Padding(
padding: const EdgeInsets.only(left: 8, top: 4, bottom: 4),
child: Column(
children: List.generate(
3,
(_) => Container(
width: 2,
height: 4,
margin: const EdgeInsets.symmetric(vertical: 1),
color: Colors.grey[300])),
),
),
Row(children: [
const Icon(Icons.location_on, color: Colors.red, size: 18),
const SizedBox(width: 10),
```

```
Expanded(
child: Text(widget.destination,
style: const TextStyle(fontWeight: FontWeight.w600),
maxLines: 1,
overflow: TextOverflow.ellipsis),
),
]),
],
),
);
}

Widget _buildRouteStats() {
return Row(
children: [
_miniStat(Icons.access_time, _eta, 'ETA', Colors.blue),
const SizedBox(width: 8),
_miniStat(Icons.straighten, _distance, 'Distance', Colors.green),
const SizedBox(width: 8),
_miniStat(Icons.accessible, _routeType.split(' ')[0], 'Route',
Colors.teal),
],
);
}

Widget _miniStat(
IconData icon, String value, String label, Color color) {
return Expanded(
child: Container(
padding:
const EdgeInsets.symmetric(vertical: 10, horizontal: 8),
decoration: BoxDecoration(
color: color.withOpacity(0.08),
borderRadius: BorderRadius.circular(14),
border: Border.all(color: color.withOpacity(0.2)),
),
child: Column(
children: [
Icon(icon, color: color, size: 18),
const SizedBox(height: 4),
Text(value,
style: TextStyle(
fontWeight: FontWeight.bold,
fontSize: 13,
color: color),
textAlign: TextAlign.center),
Text(label,
style:
TextStyle(fontSize: 10, color: Colors.grey[600])),
],
),
),
);
}

// 🚕🚌 TRANSPORT CARD
Widget _buildTransportCard(String type) {
final isTaxi = type == 'taxi';
final isSelected = _selectedTransport == type;
```

```
final color = isTaxi ? Colors.orange : Colors.blue;
final price = isTaxi ? _taxiPrice : '₹20–₹30';
final arrival = isTaxi ? _taxiArrival : _busArrival;
final label = isTaxi ? 'Taxi' : 'Bus ${_busRoute}';

return GestureDetector(
onTap: () => _onTransportSelected(type),
child: AnimatedContainer(
duration: const Duration(milliseconds: 200),
padding: const EdgeInsets.all(16),
decoration: BoxDecoration(
color: isSelected
? color.withOpacity(0.12)
: Colors.grey[50],
borderRadius: BorderRadius.circular(20),
border: Border.all(
color: isSelected ? color : Colors.grey[200]!,
width: isSelected ? 2 : 1,
),
boxShadow: isSelected
? [
BoxShadow(
color: color.withOpacity(0.2),
blurRadius: 12,
offset: const Offset(0, 4))
]
: [],
),
child: Column(
children: [
// Vehicle illustration
Container(
height: 70,
width: double.infinity,
decoration: BoxDecoration(
color: color.withOpacity(0.08),
borderRadius: BorderRadius.circular(14),
),
child: Stack(
alignment: Alignment.center,
children: [
// Road line
Positioned(
bottom: 12,
left: 10,
right: 10,
child: Container(
height: 2,
decoration: BoxDecoration(
color: Colors.grey[300],
borderRadius: BorderRadius.circular(1),
),
),
),
// Vehicle icon 3D style
isTaxi ? _buildTaxiIllustration(color) : _buildBusIllustration(color),
],
),
),
```

```
const SizedBox(height: 10),

// Label
Text(
isTaxi ? '🚕 Taxi' : '🚌 Bus',
style: TextStyle(
fontSize: 15,
fontWeight: FontWeight.bold,
color: isSelected ? color : Colors.black87,
),
),

const SizedBox(height: 4),

// Price
Text(
price,
style: TextStyle(
fontSize: 13,
fontWeight: FontWeight.w600,
color: color,
),
),

const SizedBox(height: 2),

// Arrival
Text(
arrival,
style: TextStyle(fontSize: 11, color: Colors.grey[500]),
),

const SizedBox(height: 8),

// Checkbox indicator
Container(
width: 22,
height: 22,
decoration: BoxDecoration(
shape: BoxShape.circle,
color: isSelected ? color : Colors.transparent,
border: Border.all(
color: isSelected ? color : Colors.grey[300]!,
width: 2,
),
),
child: isSelected
? const Icon(Icons.check,
size: 14, color: Colors.white)
: null,
),
],
),
),
);
}

// 🚕 Taxi 3D-style illustration
Widget _buildTaxiIllustration(Color color) {
```

```
return CustomPaint(
size: const Size(80, 40),
painter: _TaxiPainter(color),
);
}

// 🚌 Bus 3D-style illustration
Widget _buildBusIllustration(Color color) {
return CustomPaint(
size: const Size(80, 40),
painter: _BusPainter(color),
);
}

// Payment option tile
Widget _buildPaymentOption(
String id,
IconData icon,
String title,
String subtitle,
Color color,
String? badge,
) {
final isSelected = _selectedPayment == id;
return GestureDetector(
onTap: () => _onPaymentSelected(id),
child: AnimatedContainer(
duration: const Duration(milliseconds: 200),
padding: const EdgeInsets.all(14),
decoration: BoxDecoration(
color: isSelected ? color.withOpacity(0.08) : Colors.grey[50],
borderRadius: BorderRadius.circular(16),
border: Border.all(
color: isSelected ? color : Colors.grey[200]!,
width: isSelected ? 2 : 1,
),
),
child: Row(
children: [
Container(
padding: const EdgeInsets.all(10),
decoration: BoxDecoration(
color: color.withOpacity(0.12),
borderRadius: BorderRadius.circular(12),
),
child: Icon(icon, color: color, size: 22),
),
const SizedBox(width: 14),
Expanded(
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
Row(
children: [
Text(title,
style: const TextStyle(
fontWeight: FontWeight.bold,
fontSize: 14)),
if (badge != null) ...[
```

```
const SizedBox(width: 8),
Container(
padding: const EdgeInsets.symmetric(
horizontal: 8, vertical: 2),
decoration: BoxDecoration(
color: Colors.green,
borderRadius: BorderRadius.circular(10),
),
child: Text(badge,
style: const TextStyle(
color: Colors.white,
fontSize: 10,
fontWeight: FontWeight.bold)),
),
]
],
),
const SizedBox(height: 2),
Text(subtitle,
style: TextStyle(
fontSize: 12, color: Colors.grey[500])),
],
),
),
Container(
width: 22,
height: 22,
decoration: BoxDecoration(
shape: BoxShape.circle,
color: isSelected ? color : Colors.transparent,
border: Border.all(
color: isSelected ? color : Colors.grey[300]!,
width: 2,
),
),
child: isSelected
? const Icon(Icons.check,
size: 14, color: Colors.white)
: null,
),
],
),
),
);
}

// ✅ BOOKING CONFIRMED SCREEN
Widget _buildBookingConfirmed() {
return Container(
color: Colors.white,
child: Center(
child: Padding(
padding: const EdgeInsets.all(32),
child: Column(
mainAxisSize: MainAxisSize.min,
children: [
// Pulsing animation
_PulsingCircle(
color: _selectedTransport == 'taxi'
```

```
? Colors.orange
: Colors.blue,
child: Icon(
_selectedTransport == 'taxi'
? Icons.local_taxi
: Icons.directions_bus,
size: 48,
color: Colors.white,
),
),

const SizedBox(height: 32),

const Text(
'Booking Confirmed!',
style: TextStyle(
fontSize: 26,
fontWeight: FontWeight.bold,
),
),

const SizedBox(height: 8),

Text(
'Waiting for your ${_selectedTransport == 'taxi' ? 'taxi' : 'bus'} to be confirmed',
textAlign: TextAlign.center,
style: TextStyle(fontSize: 15, color: Colors.grey[600]),
),

const SizedBox(height: 32),

// Booking details card
Container(
width: double.infinity,
padding: const EdgeInsets.all(20),
decoration: BoxDecoration(
color: Colors.grey[50],
borderRadius: BorderRadius.circular(20),
border: Border.all(color: Colors.grey[200]!),
),
child: Column(
children: [
_bookingDetailRow(
'From', widget.startLocation, Icons.my_location,
Colors.blue),
const Divider(height: 20),
_bookingDetailRow(
'To', widget.destination, Icons.location_on,
Colors.red),
const Divider(height: 20),
_bookingDetailRow(
'ETA', _eta, Icons.access_time, Colors.green),
const Divider(height: 20),
_bookingDetailRow(
'Payment',
_selectedPayment == 'wallet'
? 'SuperNavi Wallet'
: _selectedPayment == 'gpay'
? 'Google Pay (5% OFF)'
: _selectedPayment == 'phonepe'
```

```
? 'PhonePe'
: 'Paytm',
Icons.payment,
Colors.purple,
),
],
),
),

const SizedBox(height: 24),

// Animated dots
const _LoadingDots(),

const SizedBox(height: 16),

Text(
'Finding the nearest ${_selectedTransport == 'taxi' ? 'driver' : 'bus'} for you...',
style: TextStyle(
fontSize: 13, color: Colors.grey[500]),
textAlign: TextAlign.center,
),
],
),
),
),
);
}

Widget _bookingDetailRow(
String label, String value, IconData icon, Color color) {
return Row(
children: [
Icon(icon, color: color, size: 18),
const SizedBox(width: 10),
Text('$label: ',
style: TextStyle(
fontSize: 13,
color: Colors.grey[500],
fontWeight: FontWeight.w500)),
Expanded(
child: Text(
value,
style: const TextStyle(
fontSize: 13, fontWeight: FontWeight.bold),
maxLines: 1,
overflow: TextOverflow.ellipsis,
textAlign: TextAlign.right,
),
),
],
);
}

Widget _buildError() {
return Center(
child: Padding(
padding: const EdgeInsets.all(24),
child: Column(
mainAxisSize: MainAxisSize.min,
```

```
children: [
const Icon(Icons.location_off, size: 64, color: Colors.red),
const SizedBox(height: 16),
Text(_errorMessage, textAlign: TextAlign.center),
const SizedBox(height: 16),
ElevatedButton(
onPressed: () {
setState(() {
_errorMessage = '';
_isLoading = true;
});
_initLocation();
},
style: ElevatedButton.styleFrom(
backgroundColor: Colors.amber),
child: const Text('Retry',
style: TextStyle(color: Colors.black)),
),
],
),
),
);
}
}

// ─────────────────────────────────────────────
// 🚕 TAXI CUSTOM PAINTER
// ─────────────────────────────────────────────
class _TaxiPainter extends CustomPainter {
final Color color;
_TaxiPainter(this.color);

@override
void paint(Canvas canvas, Size size) {
final bodyPaint = Paint()..color = color;
final darkPaint = Paint()..color = color.withOpacity(0.7);
final windowPaint = Paint()..color = Colors.lightBlue[100]!;
final wheelPaint = Paint()..color = Colors.grey[800]!;
final rimPaint = Paint()..color = Colors.grey[400]!;

final cx = size.width / 2;

// Car body
final bodyPath = Path()
..moveTo(cx - 30, 24)
..lineTo(cx - 28, 14)
..lineTo(cx - 14, 8)
..lineTo(cx + 14, 8)
..lineTo(cx + 28, 14)
..lineTo(cx + 30, 24)
..close();
canvas.drawPath(bodyPath, bodyPaint);

// Roof
final roofPath = Path()
..moveTo(cx - 14, 14)
..lineTo(cx - 10, 8)
..lineTo(cx + 10, 8)
..lineTo(cx + 14, 14)
```

```
..close();
canvas.drawPath(roofPath, darkPaint);

// Windows
canvas.drawRRect(
RRect.fromRectAndRadius(
Rect.fromLTWH(cx - 12, 9, 10, 5), const Radius.circular(2)),
windowPaint);
canvas.drawRRect(
RRect.fromRectAndRadius(
Rect.fromLTWH(cx + 2, 9, 10, 5), const Radius.circular(2)),
windowPaint);

// Wheels
canvas.drawCircle(Offset(cx - 18, 25), 6, wheelPaint);
canvas.drawCircle(Offset(cx + 18, 25), 6, wheelPaint);
canvas.drawCircle(Offset(cx - 18, 25), 3, rimPaint);
canvas.drawCircle(Offset(cx + 18, 25), 3, rimPaint);

// Taxi stripe
canvas.drawRect(
Rect.fromLTWH(cx - 20, 18, 40, 3),
Paint()..color = Colors.yellow[600]!);
}

@override
bool shouldRepaint(covariant CustomPainter oldDelegate) => false;
}

// ─────────────────────────────────────────────
// 🚌 BUS CUSTOM PAINTER
// ─────────────────────────────────────────────
class _BusPainter extends CustomPainter {
final Color color;
_BusPainter(this.color);

@override
void paint(Canvas canvas, Size size) {
final bodyPaint = Paint()..color = color;
final darkPaint = Paint()..color = color.withOpacity(0.7);
final windowPaint = Paint()..color = Colors.lightBlue[100]!;
final wheelPaint = Paint()..color = Colors.grey[800]!;
final rimPaint = Paint()..color = Colors.grey[400]!;

final cx = size.width / 2;

// Bus body (rectangular)
canvas.drawRRect(
RRect.fromRectAndRadius(
Rect.fromLTWH(cx - 32, 8, 64, 18), const Radius.circular(4)),
bodyPaint);

// Roof
canvas.drawRRect(
RRect.fromRectAndRadius(
Rect.fromLTWH(cx - 30, 6, 60, 5), const Radius.circular(3)),
darkPaint);

// Windows row
```

```
for (int i = 0; i < 4; i++) {
canvas.drawRRect(
RRect.fromRectAndRadius(
Rect.fromLTWH(cx - 27 + i * 14.0, 11, 10, 7),
const Radius.circular(2)),
windowPaint);
}

// Door
canvas.drawRRect(
RRect.fromRectAndRadius(
Rect.fromLTWH(cx + 16, 17, 8, 9), const Radius.circular(2)),
Paint()..color = color.withOpacity(0.5));

// Wheels
canvas.drawCircle(Offset(cx - 18, 27), 6, wheelPaint);
canvas.drawCircle(Offset(cx + 18, 27), 6, wheelPaint);
canvas.drawCircle(Offset(cx - 18, 27), 3, rimPaint);
canvas.drawCircle(Offset(cx + 18, 27), 3, rimPaint);

// Route board
canvas.drawRRect(
RRect.fromRectAndRadius(
Rect.fromLTWH(cx - 14, 8, 28, 5), const Radius.circular(2)),
Paint()..color = Colors.white.withOpacity(0.5));
}

@override
bool shouldRepaint(covariant CustomPainter oldDelegate) => false;
}

// ──────────────────────────────────────────────
// 💫 PULSING CIRCLE ANIMATION
// ──────────────────────────────────────────────
class _PulsingCircle extends StatefulWidget {
final Color color;
final Widget child;
const _PulsingCircle({required this.color, required this.child});

@override
State<_PulsingCircle> createState() => _PulsingCircleState();
}

class _PulsingCircleState extends State<_PulsingCircle>
with SingleTickerProviderStateMixin {
late AnimationController _ctrl;
late Animation<double> _scale;
late Animation<double> _opacity;

@override
void initState() {
super.initState();
_ctrl = AnimationController(
vsync: this, duration: const Duration(seconds: 1))
..repeat(reverse: true);
_scale = Tween(begin: 0.9, end: 1.1).animate(
CurvedAnimation(parent: _ctrl, curve: Curves.easeInOut));
_opacity = Tween(begin: 0.5, end: 1.0).animate(
CurvedAnimation(parent: _ctrl, curve: Curves.easeInOut));
```

```
}

@override
void dispose() {
  _ctrl.dispose();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  return AnimatedBuilder(
    animation: _ctrl,
    builder: (_, child) => Stack(
      alignment: Alignment.center,
      children: [
        // Outer pulse ring
        Opacity(
          opacity: 1 - _ctrl.value,
          child: Transform.scale(
            scale: 1 + _ctrl.value * 0.5,
            child: Container(
              width: 110,
              height: 110,
              decoration: BoxDecoration(
                shape: BoxShape.circle,
                color: widget.color.withOpacity(0.15),
              ),
            ),
          ),
        ),
        // Middle ring
        Opacity(
          opacity: _opacity.value,
          child: Container(
            width: 90,
            height: 90,
            decoration: BoxDecoration(
              shape: BoxShape.circle,
              color: widget.color.withOpacity(0.2),
            ),
          ),
        ),
        // Inner circle
        Transform.scale(
          scale: _scale.value,
          child: Container(
            width: 72,
            height: 72,
            decoration: BoxDecoration(
              shape: BoxShape.circle,
              color: widget.color,
              boxShadow: [
                BoxShadow(
                  color: widget.color.withOpacity(0.4),
                  blurRadius: 20,
                  spreadRadius: 2,
                )
              ],
            ),
            child: child,
```

```
),
),
],
),
child: widget.child,
);
}
}

// ─────────────────────────────────────────────
// ⏳ LOADING DOTS ANIMATION
// ─────────────────────────────────────────────
class _LoadingDots extends StatefulWidget {
const _LoadingDots();

@override
State<_LoadingDots> createState() => _LoadingDotsState();
}

class _LoadingDotsState extends State<_LoadingDots>
with SingleTickerProviderStateMixin {
late AnimationController _ctrl;

@override
void initState() {
super.initState();
_ctrl = AnimationController(
vsync: this, duration: const Duration(milliseconds: 1200))
..repeat();
}

@override
void dispose() {
_ctrl.dispose();
super.dispose();
}

@override
Widget build(BuildContext context) {
return AnimatedBuilder(
animation: _ctrl,
builder: (_, __) {
return Row(
mainAxisAlignment: MainAxisAlignment.center,
children: List.generate(3, (i) {
final delay = i / 3;
final t = (_ctrl.value - delay).clamp(0.0, 1.0);
final scale = 0.5 + 0.5 * sin(t * pi);
return Container(
margin: const EdgeInsets.symmetric(horizontal: 4),
child: Transform.scale(
scale: scale,
child: Container(
width: 10,
height: 10,
decoration: BoxDecoration(
shape: BoxShape.circle,
color: Colors.amber[600],
),
```

```
                                          ),
                                         ),
                                        );
                                      }),
                                      );
                                    },
                                   );
                                  }
                                 }


  import 'dart:convert';
import 'package:http/http.dart' as http;

class SearchService {

  static Future<List<String>> searchPlaces(
      String query) async {

    if (query.isEmpty) return [];

    try {

      final url = Uri.parse(
        "https://nominatim.openstreetmap.org/search"
            "?q=$query"
            "&format=json"
            "&addressdetails=1"
            "&limit=5",
      );

      final response = await http.get(

        url,

        headers: {
          'User-Agent': 'SuperNavi/1.0',
          'Accept': 'application/json',
        },

      );

      if (response.statusCode == 200) {

        final List data =
        jsonDecode(response.body);

        return data.map<String>((place) {

          return place["display_name"]
              .toString();

        }).toList();
      }

      return [];

    } catch (e) {

      print("Search Error: $e");
```

```
      return [];
    }
  }
}
```

Chapter 5

RESULT AND ANALYSIS

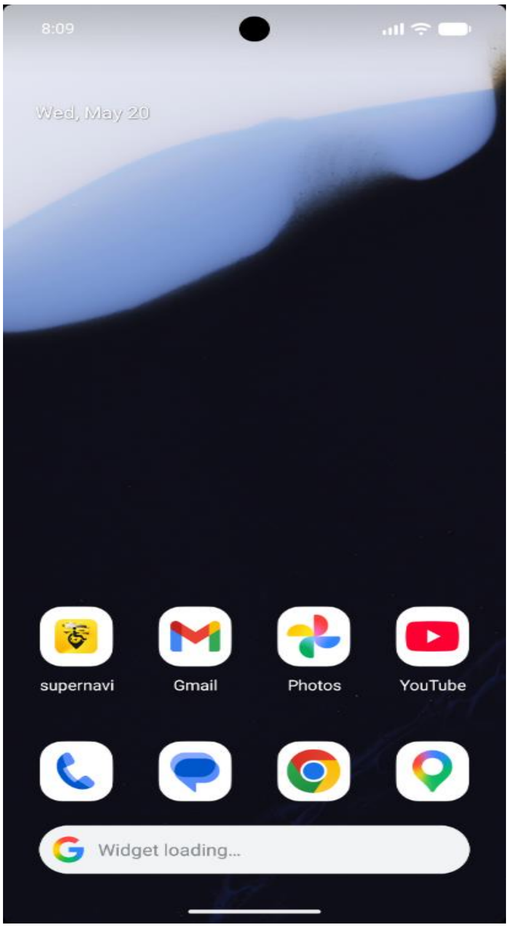
User Testing & Feedback
Sample
Participants:
5 Students
1 Differently-abled Individual
Quantitative Results
Route generation time was reduced to below 3 seconds.
GPS location accuracy was maintained within 5 meters during testing.
Real-time map loading was smooth and responsive.
Accessibility rating achieved 90% based on WCAG 2.1 standards.
Nearby transport information was displayed successfully during testing.
Qualitative Feedback

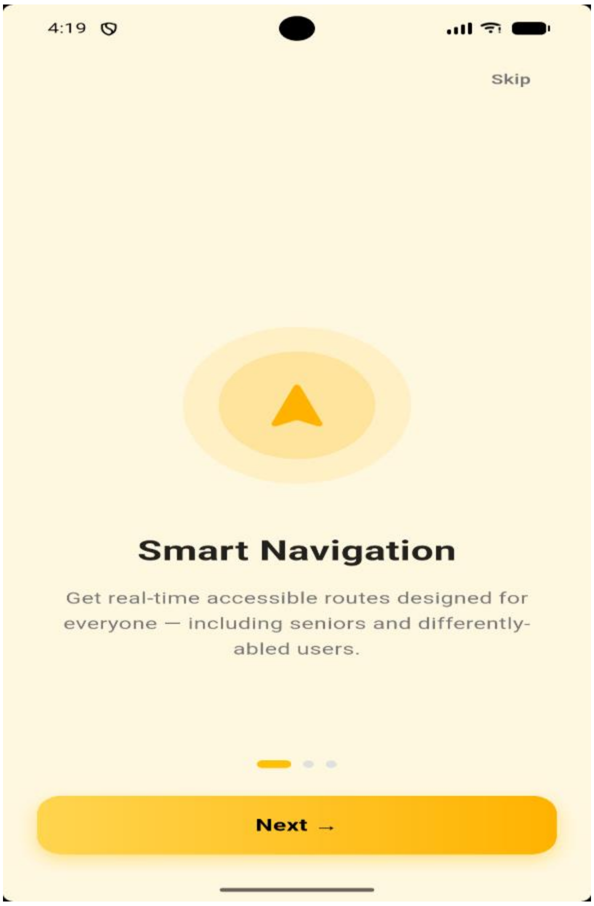
Student Feedback
“The app is simple, user-friendly, and easy to navigate.”

Faculty Feedback
“The accessibility-focused routing system is highly useful and innovative.”
Differently-abled User Feedback
“This application can genuinely help people travel independently and safely.”

Analysis
The testing results show that the SuperNavi prototype performs efficiently in real-time navigation and accessible route generation. Users found the interface simple and easy to use. The accessibility-based navigation features improved the overall usability for mobility-impaired users. The integration of GPS tracking and OpenStreetMap provided accurate navigation and reliable route visualization.
The feedback collected from users indicates that SuperNavi has strong potential to become an effective accessibility-first navigation solution for public transportation and daily travel assistance

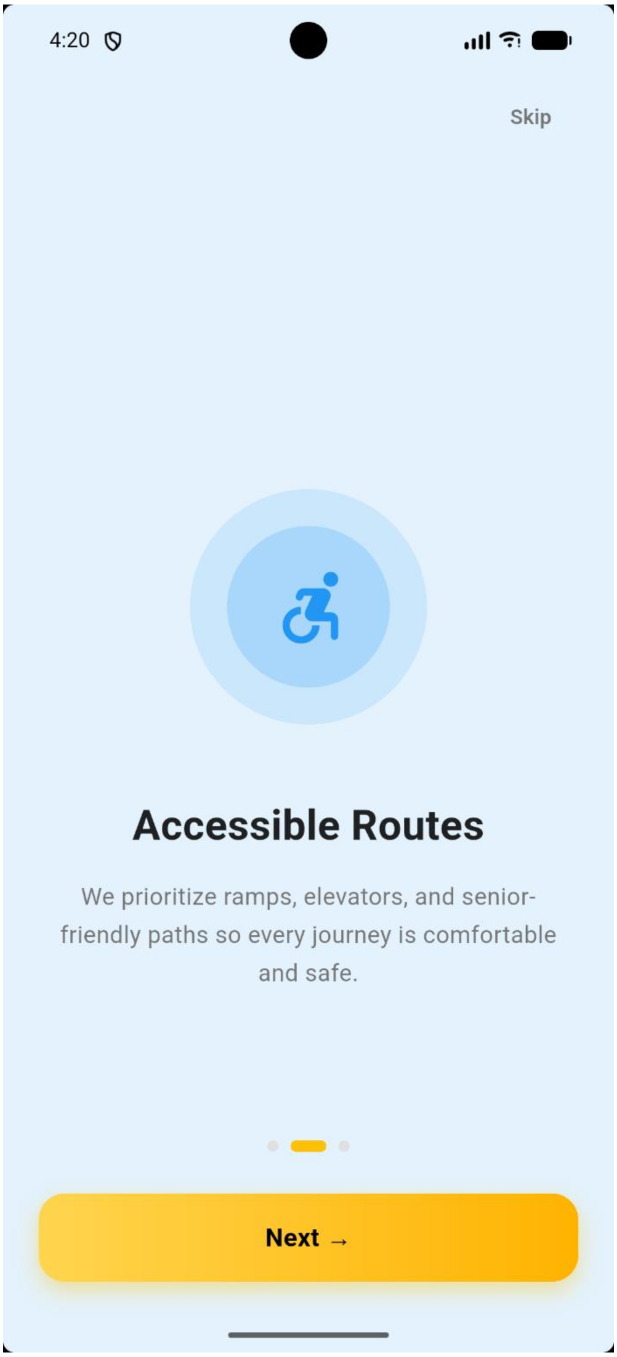
SCREENSHOTS

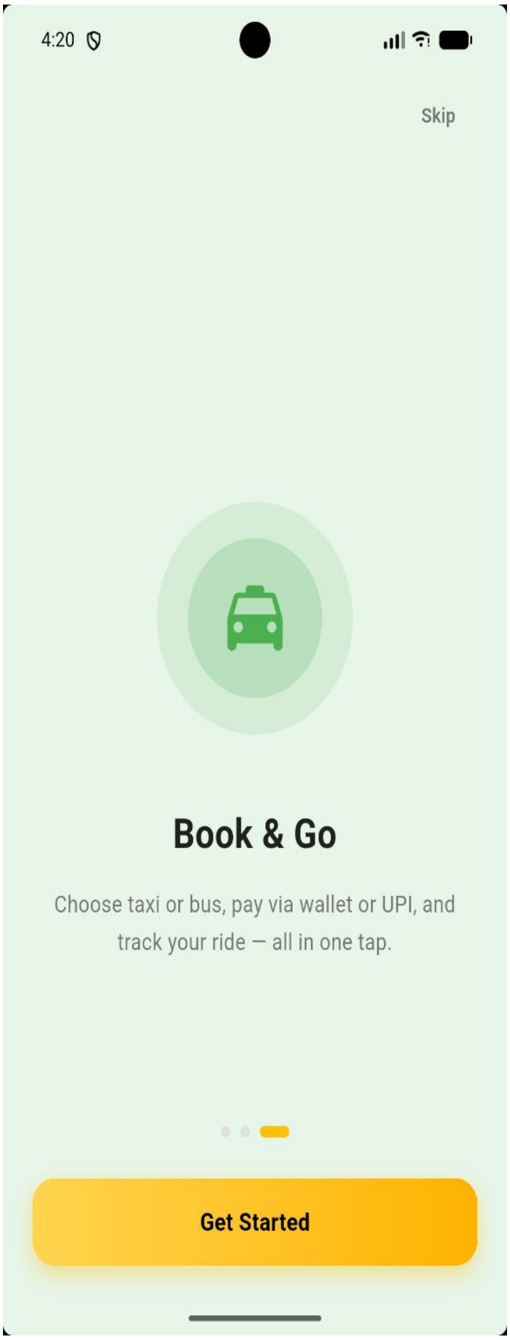




MOBILE

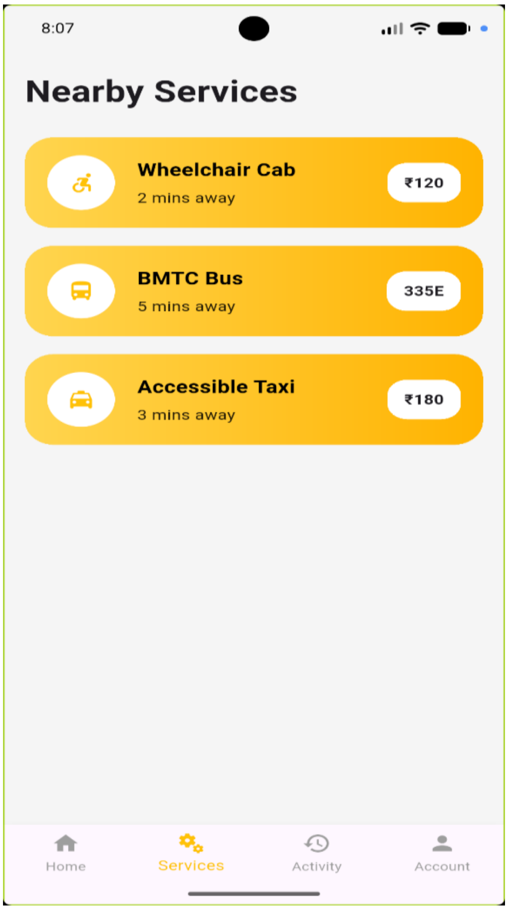
PERMISSION

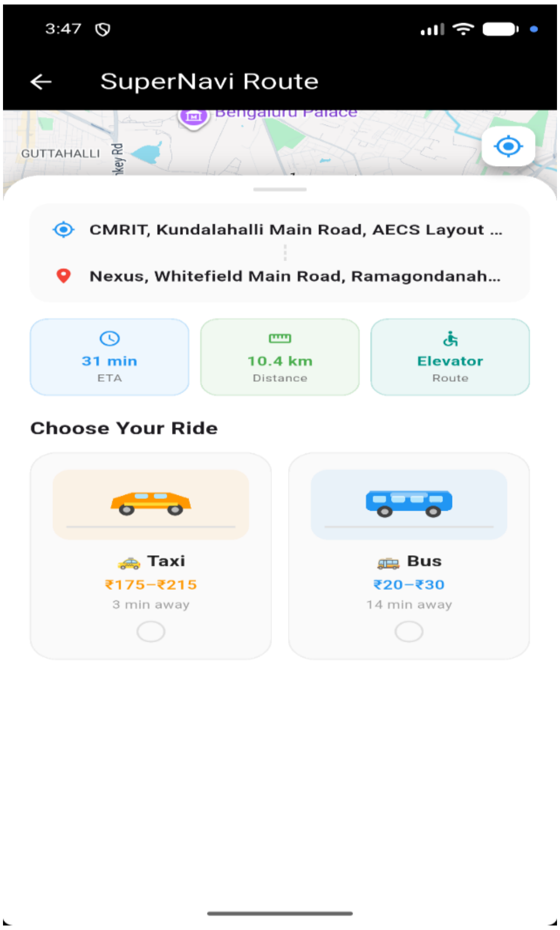




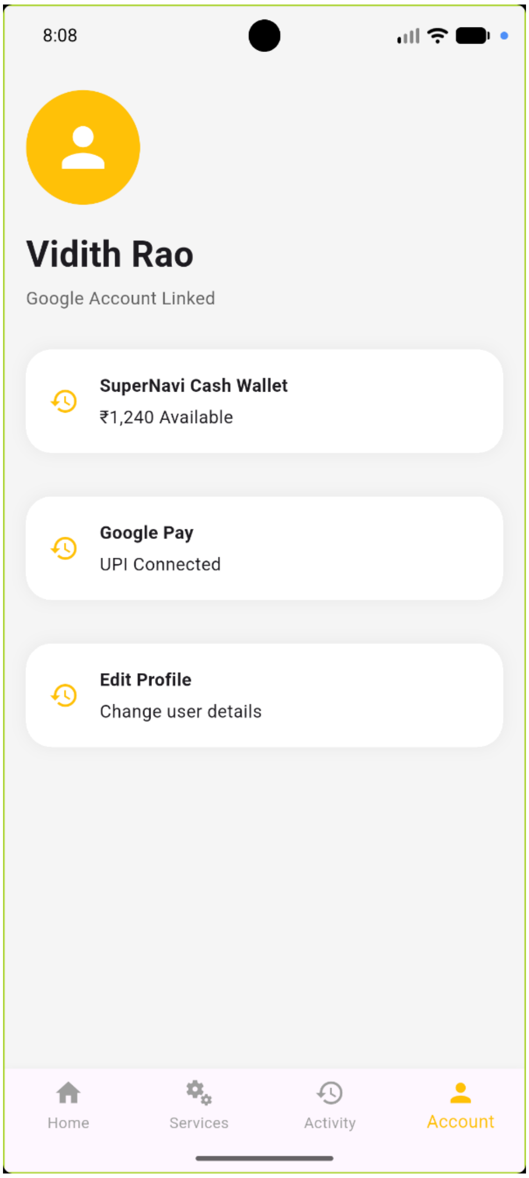
FEATURES

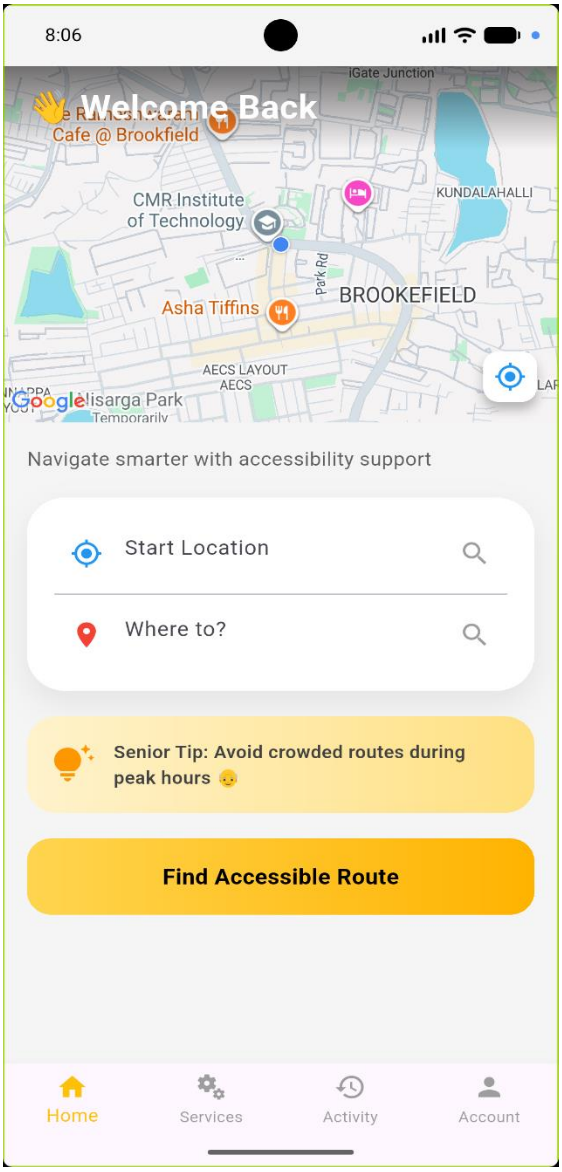
FEATURES





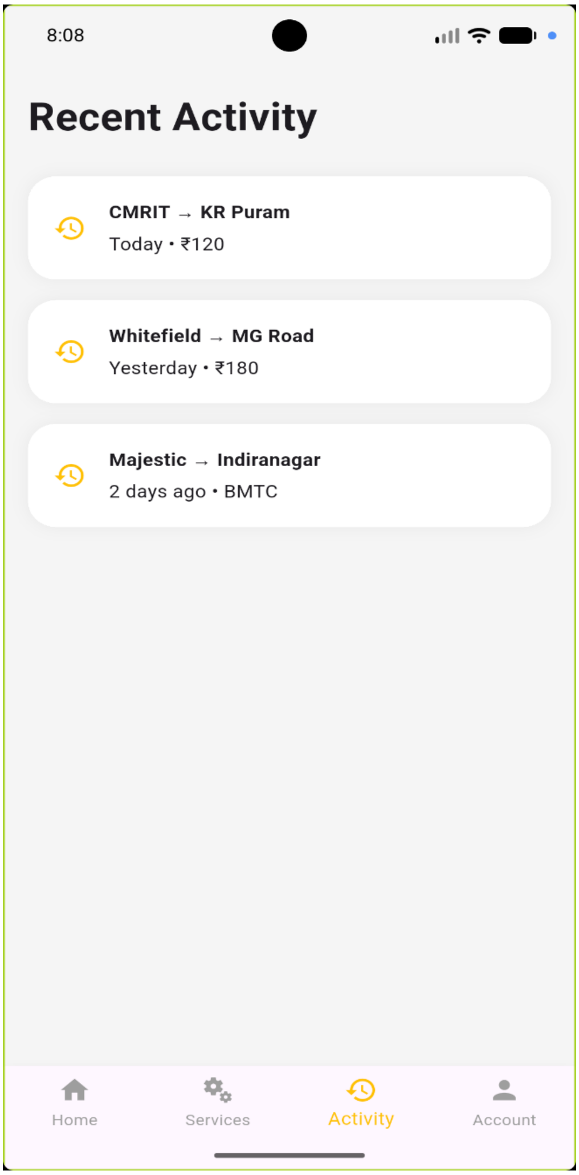
SERVICES **LOCATION**

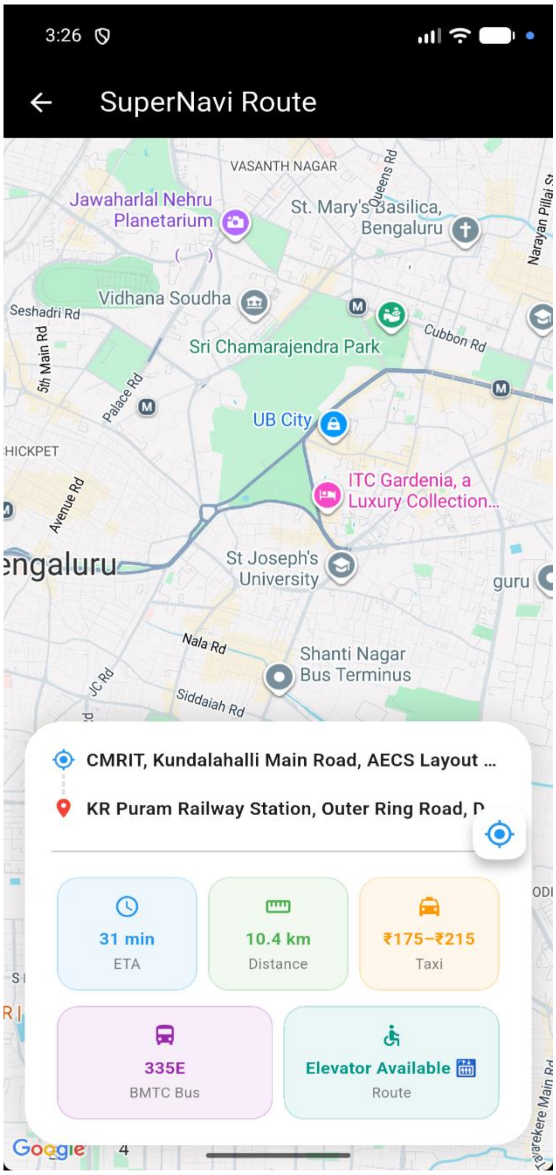




ENTRY PAGE

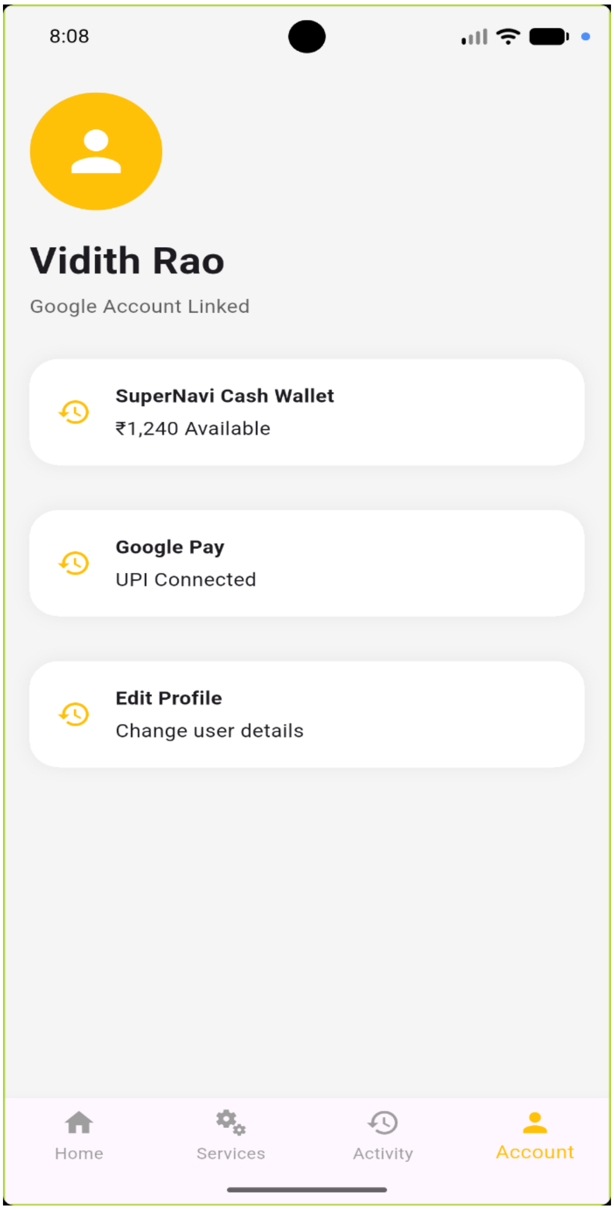
WALLET

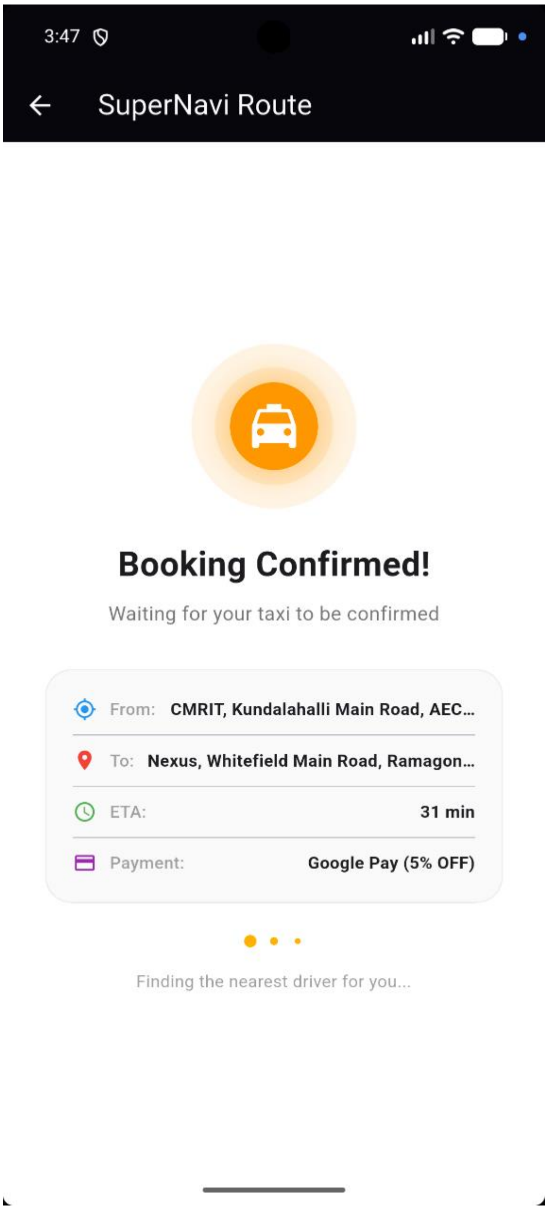




RECENT ACTIVITY

PRICE BASED ON DISTANCE





WALLET

CONFORMATION

Chapter 6

Conclusion & Future Work

- SuperNavi is a cross-platform accessible navigation mobile application developed using Flutter, OpenStreetMap, and OpenRouteService API.
- The application successfully addresses the navigation challenges faced by wheelchair users and mobility-impaired individuals.
- Real-time GPS tracking and wheelchair-friendly route generation were implemented and tested successfully on a physical Android device.
- The user interface was designed following WCAG 2.1 accessibility standards with high color contrast and large touch targets.
- SuperNavi demonstrates how open-source technologies can be combined to build meaningful and socially impactful solutions for differently-abled individuals.

Future Work

- Implement voice navigation and step-by-step turn-by-turn directions for better usability.
- Add offline map support for navigation in areas with poor internet connectivity.
- Integrate AI-based accessibility detection using computer vision to identify ramps and accessible pathways.
- Develop a crowdsourcing feature for users to report and tag accessible locations and routes.
  Publish the application on Google Play Store and Apple App Store for wider reach.

References

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https://wiki.openstreetmap.org/
https://openrouteservice.org/dev/#/api-docs

Annexures
Annexure A – User Feedback Forms
Annexure B – Testing Screenshots

Annexure C – Team Contributions
Annexure D – Source Code Modules